

Lecture 7: Poisson distribution continued & Continuous distributions

BIOS 600 - Fall 2025

UNC - Chapel Hill

Announcements

- ▶ HW 2 due tonight at 11:59pm
- ▶ Application Exercise due tomorrow (Friday) at 11:59pm
- ▶ HW 3 (short) will be released after class, due next Thursday at 11:59pm.
- ▶ Today's lecture marks the end of the content for Exam 01! (Today's lecture is included in Exam 01 content.)
- ▶ Topics list and Formula sheet have been added to Exam 01 Preparation on Canvas.

Announcements

- ▶ No new labs on 9/16 and 9/23 as we lead up to the exam. These lab sessions will be “TA Office Hours” - time for you to get help on any class material.
- ▶ If you have any questions, feel free to make a “New Post” on Ed Discussion about it! We are happy to answer any questions, no matter how small.

Overview

- ▶ Review discrete probability distributions
- ▶ Poisson distribution
- ▶ Work on Application Exercise from last time
- ▶ Expectation and variance
- ▶ Continuous distributions and examples

Supplemental Reading

- ▶ Pagano and Gavreau: Section 7.4
- ▶ [OpenIntro Statistics](#): 3.5, 4.1

Review: Discrete probability distributions

There are three rules for discrete probability distributions:

1. Outcomes must be disjoint
2. The probability of each outcome must be ≥ 0 and ≤ 1
3. The sum of the outcome probabilities must add up to 1

Event	Probability
$X = pre$	0.10
$X = early$	0.27
$X = full$	0.57
$X = late/post$	0.06

The Poisson distribution

- ▶ Discrete distribution taking on possible values 0, 1, 2, ..., ∞
- ▶ Often used to model counts or rare events
- ▶ Much like the binomial distribution, requires a few assumptions



The Poisson distribution

The **Poisson distribution** gives the probability that k events occur in a given “interval”. There are four assumptions:

1. Within any interval, k may take on values $0, 1, 2, 3, \dots, \infty$
2. Each event occurs independently, both within the same interval, and between intervals
3. The average rate at which events occur in an interval, λ , is constant
4. Two events cannot occur simultaneously

What is an “interval”?

The Poisson distribution

If X has a Poisson distribution, then

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

What do these components mean?

i Note

This expression will always be provided to you if needed.

Poisson Example: Alzheimer's Disease

Suppose on average, there are 1.5 deaths due to Alzheimer's Disease in a town each year. For a one-year period in this town, what is the chance that two or more people die from Alzheimer's?

- ▶ Let's verify that our example follows the four assumptions of a Poisson distribution.

Verify Assumptions

1. Within any interval, the number of AD deaths can range from 0 to ∞ (*technically* not true, but close enough)
2. One individual dying of Alzheimer's does not affect the chance of another person dying of Alzheimer's
3. The AD death rate is constant in this town
4. Two AD deaths cannot occur at the same time (we can always subdivide time intervals such that only one person experiences this event in a given sub-interval)

Example



Practice with Poisson

Using the formula for a Poisson distribution, write out how you would calculate the probability of zero Alzheimer's deaths in a year.

The Poisson distribution

- ▶ Thus $Z \sim \text{Pois}(1.5)$, and $P(Z = 0)$, $P(Z = 1)$, and $P(Z \geq 2)$ are disjoint events.
- ▶ So, using the Complement Rule,
 $P(Z \geq 2) = 1 - [P(Z = 0) + P(Z = 1)]$, where

$$P(Z = 0) = \frac{1.5^0 \times e^{-1.5}}{0!} \approx 0.223$$

$$P(Z = 1) = \frac{1.5^1 \times e^{-1.5}}{1!} \approx 0.335$$

And so $P(Z \geq 2) \approx 1 - 0.223 - 0.335 = 0.442$.

In R

We can calculate these values in R using the `dpois()` function:

```
dpois(x = 0, lambda = 1.5)
```

```
[1] 0.2231302
```

```
dpois(x = 1, lambda = 1.5)
```

```
[1] 0.3346952
```

```
1- (dpois(x = 0, lambda = 1.5) +  
     dpois(x = 1, lambda = 1.5))
```

```
[1] 0.4421746
```

Note: Each parentheses needs a buddy! Common source of Quarto files not rendering...

In R

If we wanted to abbreviate our typing (and reduce possibility of typos), we can save our first two probabilities as objects `p0` and `p1` to use them later.

```
p0 <- dpois(x = 0, lambda = 1.5)
p0
```

```
[1] 0.2231302
```

```
p1 <- dpois(x = 1, lambda = 1.5)
p1
```

```
[1] 0.3346952
```

```
1- (p0 + p1)
```

```
[1] 0.4421746
```

Participation

Application Exercise in R

In today's participation, continue with Exercise 3 in Application Exercise 01 (from last time).

When you've completed the exercises, render, save to PDF, and submit on Canvas> Assignments. Due **Friday at 11:59pm**.

What about other interval lengths?

Suppose we have a count random variable that follows a Poisson distribution:

- ▶ Since each event is independent of others and the rate λ is constant, **the probability that an event occurs within an interval is proportional to the length of that interval.**
- ▶ E.g., we would expect twice the number of events to occur in an interval of twice the length; we would expect $1/9$ times the number of events to occur in an interval 9 times as small; and so on.

What about other interval lengths?

Suppose on average, there are 1.5 deaths due to Alzheimer's disease in a town each year.



Example:

1. What is the average one-**month** rate of deaths due to Alzheimer's disease in this town?
2. For any given one-**month** period in this town, what is the probability that exactly one person dies from Alzheimer's? (Just the expression is fine)

For $X \sim \text{Pois}(\lambda)$,

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Expectation and variance

Now that we've defined random variables and explored a few distributions, we might be interested in some other aspects of their distributions.

Suppose we are interested in the probability distribution corresponding to the number of preterm births in a random sample of five independent US births.

- ▶ How many preterm births should we expect?
- ▶ How “spread out” would this distribution be?

Expected value

The **expected value** of a discrete random variable X is a weighted average of the possible outcomes:

$$E(X) = \sum_{\text{all } x} x \cdot P(X = x)$$

This is a property of the **distribution** of X , not of the random variable itself.

Properties of Bernoulli and binomial random variables

► For $Z \sim \text{Bern}(p)$, $E(Z) = p$ and $\text{Var}(Z) = p(1 - p)$

Note

These formulas will always be provided to you if needed.

Example

Let Z be a coin toss. What is $E(Z)$ and $\text{Var}(Z)$?

$$Z \sim \text{Bern}(.5).$$

$$E(Z) = .5$$

$$\text{Var}(Z) = 0.5 \cdot (0.5) = 0.25.$$

Properties of binomial and Poisson random variables

- ▶ For $Z \sim \text{Binom}(n, p)$, $E(Z) = np$ and $\text{Var}(Z) = np(1 - p)$
- ▶ For $Z \sim \text{Pois}(\lambda)$, $E(Z) = \text{Var}(Z) = \lambda$

i Note

These formulas will always be provided to you if needed.

Why are these important?

- ▶ The expectation is the average value (weighted by the probability of each value occurring)
- ▶ The variance describes the expected spread of values around the population expectation (thus, variance is in fact an expectation itself!)

Continuous distributions

Can we be more precise?

Letting X be the random variable that corresponds to how long a baby's gestation was, we could imagine subdividing further and further:

Event	Probability
$X < 20$ wk.	$P(X < 20)$
$X = 20$ to 21 wk.	etc.
$X = 21$ to 22 wk.	etc.
$X = 22$ to 23 wk.	etc.
$\vdots$	$\vdots$

Event	Probability
$X < 20$ wk.	$P(X < 20)$
$X = 20$ to 20.1 wk.	etc.
$X = 20.1$ to 20.2 wk.	etc.
$X = 20.2$ to 20.3 wk.	etc.
$\vdots$	$\vdots$

Can we be more precise?

- ▶ Now let gestational age X be a **continuous** random variable, which can take on *any* value, say from 0 to ∞ .
- ▶ How might we define a continuous probability distribution that corresponds to X ?

Continuous probability distributions

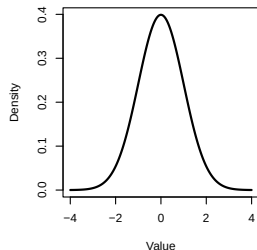
- ▶ The probability that a continuous variable equals any specific value is 0
- ▶ No use tabulating - there is an *uncountably* infinite number of possible values they can be, all with $P(X = x) = 0$
- ▶ The distribution is given by a **probability density function**, helps us describe probabilities for *ranges* of values.

Density functions

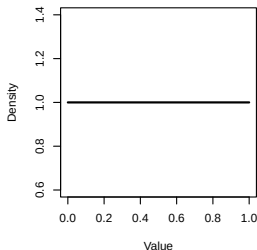
Probability density functions may be given graphically, satisfying the following two rules:

- ▶ The density must be non-negative everywhere ($f(x) \geq 0$ for all x from $-\infty$ to ∞)
- ▶ The total area under the density must be 1

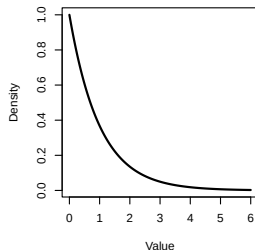
Normal Distribution



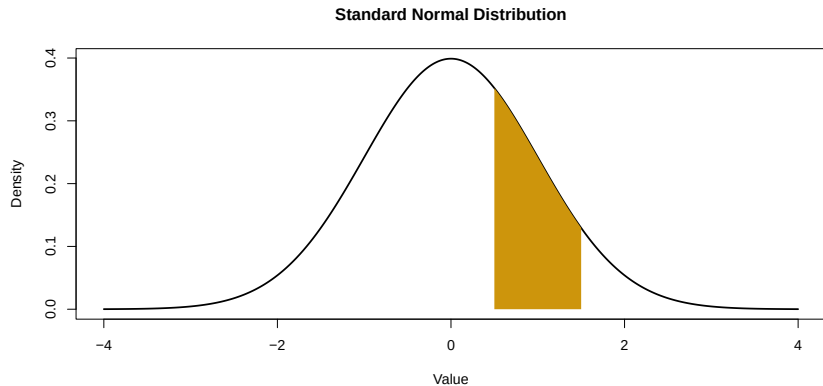
Uniform Distribution



Exponential Distribution



With density functions, we work with areas under the curve



The normal (Gaussian) distribution

For the **normal distribution**,

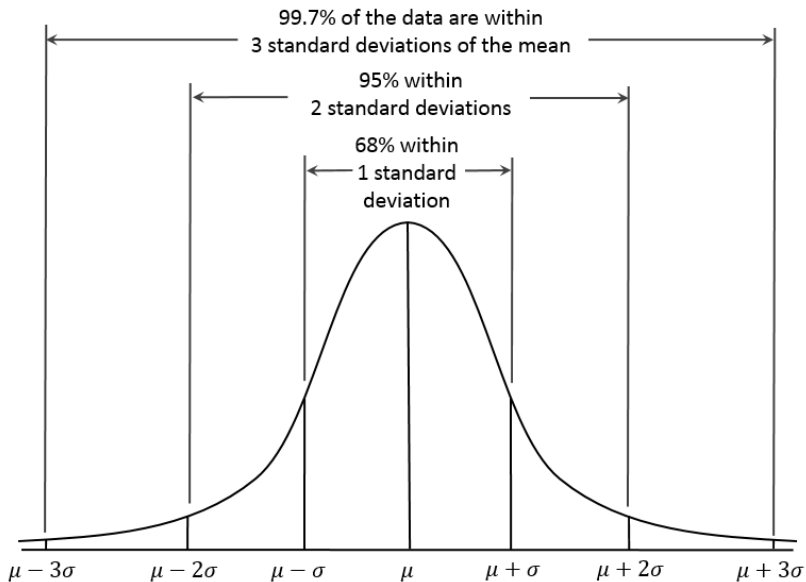
You do NOT need to know this for

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2} \frac{(x - \mu)^2}{\sigma^2}\right\}$$

where μ is the mean and σ^2 is the variance.

► We often write $N(\mu, \sigma^2)$.

68-95-99.7



Standardization

- ▶ The normal distribution is a family of distributions of a specific form. There are an infinite amount of possible distributions, since μ can be any real number and σ^2 can be any positive number.
- ▶ It would be very cumbersome to have to individually think about a $N(0, 20)$ vs. $N(2.5, 2)$ vs. $N(694, 1549)$ vs. distribution, depending on the situation.
- ▶ In practice, we could calculate a **standard score** or **z-score** that gives the number of standard deviations away from the mean an observation from a particular population is.

z-scores

- ▶ A **z-score** tells us how many population standard deviations an observation is away from the population mean.
- ▶ They provide ways to compare results across many different measurement scales, since z-scores are *unitless*

$$z = \frac{x - \mu}{\sigma}$$

(note the use of population parameters μ and σ)

- ▶ So, a z-score of 1.2 is 1.2 standard deviations above a mean; a z-score of -0.8 is 0.8 standard deviations below the mean.

Osteoporosis

- ▶ According to NHANES, the mean bone mineral density for a 65 year old white woman is 809 mg/cm^2 , with a standard deviation of 140 mg/cm^2 .
- ▶ Suppose you are a 65 year old white woman whose bone density is 698 mg/cm^2 .



Should you be very concerned about osteoporosis?
Let's calculate a z-score and find out!

Recap

- ▶ Review discrete distributions
- ▶ Poisson distribution and Alzheimer's example
- ▶ Continuous distributions, density functions
- ▶ The normal (Gaussian) distribution
- ▶ z-scores

Next class

- ▶ The Central Limit Theorem